WEB TECHNOLOGIES ASSIGNMENT – 1

REGISTER NUMBER: 953624104176

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56. Waste Collection Management System

A city project plans to develop a Waste Collection Management System that enables residents to schedule waste pickups, report uncollected waste, and monitor collection schedules online. Design and develop a responsive front-end prototype using HTML5, CSS3, and JavaScript. Include waste categories, pickup scheduling, complaint forms, and collection status pages. Use JavaScript to validate addresses, pickup dates, and complaint details.

CODE:

Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Waste Collection Management</title>
   <link rel="stylesheet" href="style.css">
</head>
<body>
<header>
   <h1>Waste Collection Management System</h1>
   <p>Clean City, Green City</p>
</header>
<nav>
   <a href="index.html">Home</a>
   <a href="schedule.html">Schedule Pickup</a>
   <a href="complaint.html">Complaint</a>
</nav>
<main>
   <section class="hero">
      <h2>Welcome to Waste Collection Management</h2>
      <p>
         Schedule your waste pickup and report uncollected waste
```

```
        easily through our online system.
    </p>
</section>
<section>
    <h2>Waste Categories</h2>
    <div class="cards">
        <div class="card">
            <h3>♻️ Dry Waste</h3>
            <p>Paper, plastic, cardboard and glass.</p>
        </div>
        <div class="card">
            <h3>🥔 Wet Waste</h3>
            <p>Food waste and other organic materials.</p>
        </div>
        <div class="card">
            <h3>🔋 E-Waste</h3>
            <p>Batteries, computers and electronic items.</p>
        </div>
    </div>
</section>
<section>
    <h2>Collection Status</h2>
    <section>
<h2>Collection Status</h2>
<div class="status">
    <h3>Morning Route</h3>
    <p id="morningStatus" class="completed">Completed ✅</p>
</div>
<div class="status">
    <h3>Afternoon Route</h3>
    <p id="afternoonStatus" class="progress">In Progress 🚚</p>
</div>
<div class="status">
```

```
        <h3>Evening Route</h3>
        <p id="eveningStatus" class="scheduled">Scheduled 🕒</p>
    </div>
    <div class="form-container">
        <h2>Edit Collection Status</h2>
        <label>Select Route</label>
        <select id="route">
            <option value="">Select Route</option>
            <option value="morning">Morning Route</option>
            <option value="afternoon">Afternoon Route</option>
            <option value="evening">Evening Route</option>
        </select>
        <label>Select Status</label>
        <select id="collectionStatus">
            <option value="">Select Status</option>
            <option value="scheduled">Scheduled</option>
            <option value="progress">In Progress</option>
            <option value="completed">Completed</option>
            <option value="notcollected">Not Collected</option>
        </select>
        <button onclick="updateStatus()">
            Update Status
        </button>
        <p id="statusMessage"></p>
    </div>
</section>
    </section>
</main>
<footer>
    <p>© 2026 Waste Collection Management System</p>
</footer>
</body>
</html>
```

Schedule.html

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Schedule Pickup</title>
   <link rel="stylesheet" href="style.css">
</head>
<body>
<header>
   <h1>Schedule Waste Pickup</h1>
</header>
<nav>
   <a href="index.html">Home</a>
   <a href="schedule.html">Schedule Pickup</a>
   <a href="complaint.html">Complaint</a>
</nav>
<main>
   <section class="form-container">
      <h2>Pickup Request</h2>
      <form id="pickupForm">
         <label>Resident Name</label>
         <input type="text" id="name"
               placeholder="Enter your name">
         <label>Address</label>
         <input type="text" id="address"
               placeholder="Enter your address">
         <label>Waste Category</label>
         <select id="waste">
            <option value="">Select Category</option>
            <option value="Dry Waste">Dry Waste</option>
            <option value="Wet Waste">Wet Waste</option>
```

```
            <option value="E-Waste">E-Waste</option>
        </select>
        <label>Pickup Date</label>
        <input type="date" id="date">
        <button type="submit">
            Schedule Pickup
        </button>
        <p id="pickupMessage"></p>
    </form>
  </section>
</main>
<footer>
  <p>© 2026 Waste Collection Management System</p>
</footer>
<script src="script.js"></script>
</body>
</html>
```

Complaint.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Report Complaint</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
<header>
  <h1>Report Uncollected Waste</h1>
</header>
<nav>
  <a href="index.html">Home</a>
  <a href="schedule.html">Schedule Pickup</a>
```

```
    <a href="complaint.html">Complaint</a>
</nav>
<main>
    <section class="form-container">
        <h2>Complaint Form</h2>
        <form id="complaintForm">
            <label>Resident Name</label>
            <input type="text" id="complaintName"
                placeholder="Enter your name">
            <label>Address</label>
            <input type="text" id="complaintAddress"
                placeholder="Enter waste collection location">
            <label>Complaint Details</label>
            <textarea id="complaint"
                placeholder="Describe the problem"></textarea>

            <label>Waste Category</label>
            <select id="complaintWaste">
                <option value="">Select Category</option>
                <option value="Dry Waste">Dry Waste</option>
                <option value="Wet Waste">Wet Waste</option>
                <option value="E-Waste">E-Waste</option>
            </select>
            <button type="submit">
                Submit Complaint
            </button>
            <p id="complaintMessage"></p>
        </form>
    </section>
</main>
<footer>
    <p>© 2026 Waste Collection Management System</p>
</footer>
```

```
<script src="script.js"></script>
</body>
</html>
```

Style.css

```
* {
    box-sizing: border-box;
    margin: 0;
    padding: 0;
}
body {
    font-family: Arial, sans-serif;
    background-color: #f4f8f4;
    color: #333;
}
header {
    background-color: #2e7d32;
    color: white;
    text-align: center;
    padding: 25px;
}
header h1 {
    margin-bottom: 8px;
}
nav {
    background-color: #1b5e20;
    text-align: center;
    padding: 15px;
}
nav a {
    color: white;
    text-decoration: none;
    margin: 0 20px;
    font-weight: bold;
```

```
}
nav a:hover {
    text-decoration: underline;
}
main {
    width: 90%;
    max-width: 1000px;
    margin: 30px auto;
}
section {
    margin-bottom: 35px;
}
section h2 {
    color: #2e7d32;
    margin-bottom: 20px;
}
.hero {
    background-color: white;
    padding: 30px;
    text-align: center;
    border-radius: 10px;
    box-shadow: 0 2px 8px #ccc;
}
.hero h2 {
    margin-bottom: 10px;
}
.cards {
    display: flex;
    gap: 20px;
    flex-wrap: wrap;
}
.card {
    background-color: white;
```

```
    padding: 25px;
    flex: 1;
    min-width: 200px;
    text-align: center;
    border-radius: 10px;
    box-shadow: 0 2px 8px #ccc;
}
.card h3 {
    color: #2e7d32;
    margin-bottom: 10px;
}
.status {
    background-color: white;
    padding: 18px;
    margin-bottom: 12px;
    border-radius: 8px;
    box-shadow: 0 2px 5px #ddd;
}
.completed {
    color: green;
}
.progress {
    color: orange;
}
.scheduled {
    color: blue;
}
.form-container {
    background-color: white;
    max-width: 600px;
    margin: auto;
    padding: 30px;
    border-radius: 10px;
```

```
    box-shadow: 0 2px 8px #ccc;
}
label {
    display: block;
    margin-top: 15px;
    font-weight: bold;
}
input,
select,
textarea {
    width: 100%;
    padding: 12px;
    margin-top: 6px;
    border: 1px solid #aaa;
    border-radius: 5px;
    font-size: 15px;
}
textarea {
    height: 120px;
    resize: none;
}
button {
    width: 100%;
    padding: 12px;
    margin-top: 20px;
    background-color: #2e7d32;
    color: white;
    border: none;
    border-radius: 5px;
    font-size: 16px;
    cursor: pointer;
}
button:hover {
```

```
    background-color: #1b5e20;
}
.success {
    color: green;
    font-weight: bold;
    margin-top: 15px;
}
.error {
    color: red;
    font-weight: bold;
    margin-top: 15px;
}
footer {
    background-color: #1b5e20;
    color: white;
    text-align: center;
    padding: 18px;
    margin-top: 40px;
}
@media (max-width: 600px) {
    nav a {
        display: block;
        margin: 10px;
    }
    .cards {
        flex-direction: column;
    }
    main {
        width: 95%;
    }
}
.status .error {
    color: red;
```

```
}
#statusMessage {
   margin-top: 15px;
}
```

Script.js

```
let pickupForm = document.getElementById("pickupForm");
if (pickupForm) {
   pickupForm.addEventListener("submit", function(event) {
      event.preventDefault();
      let name = document.getElementById("name").value.trim();
      let address = document.getElementById("address").value.trim();
      let waste = document.getElementById("waste").value;
      let date = document.getElementById("date").value;
      let message = document.getElementById("pickupMessage");
      if (name === "") {
         message.innerHTML = "Please enter your name.";
         message.className = "error";
         return;
      }
      if (address.length < 5) {
         message.innerHTML = "Please enter a valid address.";
         message.className = "error";
         return;
      }
      if (waste === "") {
         message.innerHTML = "Please select a waste category.";
         message.className = "error";
         return;
      }
      if (date === "") {
         message.innerHTML = "Please select a pickup date.";
         message.className = "error";
         return;
```

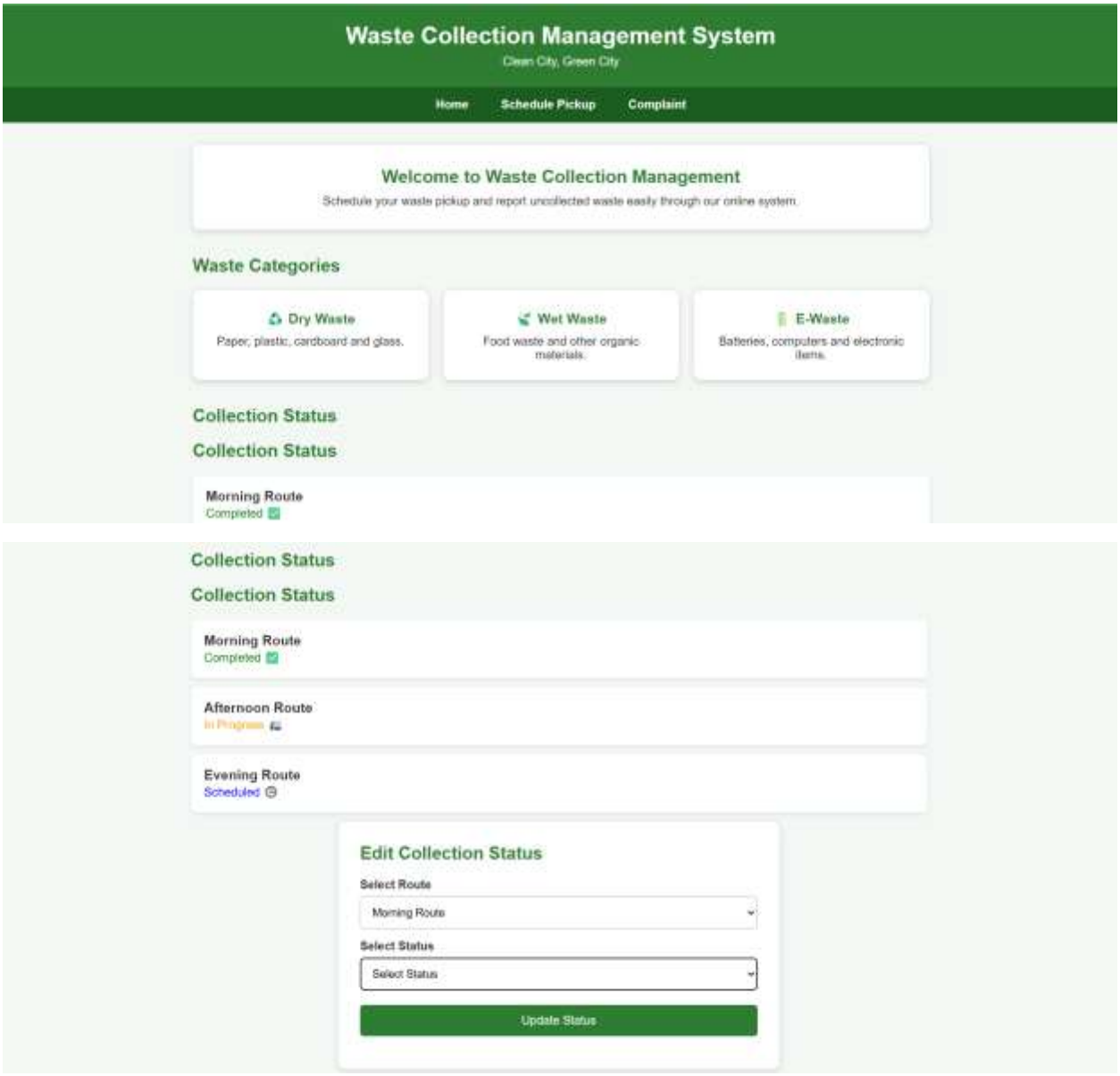
```
        }
        let today = new Date();
        today.setHours(0, 0, 0, 0);
        let selectedDate = new Date(date);
        if (selectedDate < today) {
            message.innerHTML =
                "Pickup date cannot be in the past.";
            message.className = "error";
            return;
        }
        message.innerHTML =
            "Pickup scheduled successfully! 🚚";
        message.className = "success";
        pickupForm.reset();
    });
}
let complaintForm =
    document.getElementById("complaintForm");
if (complaintForm) {
    complaintForm.addEventListener("submit", function(event) {
        event.preventDefault();
        let name =
            document.getElementById("complaintName").value.trim();
        let address =
            document.getElementById("complaintAddress").value.trim();
        let complaint =
            document.getElementById("complaint").value.trim();
        let waste =
            document.getElementById("complaintWaste").value;
        let message =
            document.getElementById("complaintMessage");
        if (name === "") {
            message.innerHTML =
```

```
            "Please enter your name.";
            message.className = "error";
            return;
        }
        if (address.length < 5) {
            message.innerHTML =
                "Please enter a valid address.";
            message.className = "error";
            return;
        }
        if (complaint.length < 10) {
            message.innerHTML =
                "Please provide more complaint details.";
            message.className = "error";
            return;
        }
        if (waste === "") {
            message.innerHTML =
                "Please select a waste category.";
            message.className = "error";
            return;
        }
        message.innerHTML =
            "Complaint submitted successfully! ✅";
        message.className = "success";
        complaintForm.reset();
    });
}
function updateStatus() {
    let route = document.getElementById("route").value;
    let status = document.getElementById("collectionStatus").value;
    let message = document.getElementById("statusMessage");
    if (route === "") {
```

```
    message.innerHTML = "Please select a route.";
    message.className = "error";
    return;
}
if (status === "") {
    message.innerHTML = "Please select a status.";
    message.className = "error";
    return;
}
let statusText = "";
let statusClass = "";
if (status === "scheduled") {
    statusText = "Scheduled 🕒";
    statusClass = "scheduled";
}
else if (status === "progress") {
    statusText = "In Progress 🚚";
    statusClass = "progress";
}
else if (status === "completed") {
    statusText = "Completed ✔";
    statusClass = "completed";
}
else if (status === "notcollected") {
    statusText = "Not Collected ✖";
    statusClass = "error";
}
let statusElement;
if (route === "morning") {
    statusElement = document.getElementById("morningStatus");
}
else if (route === "afternoon") {
    statusElement = document.getElementById("afternoonStatus");
```

```
    }
    else if (route === "evening") {
        statusElement = document.getElementById("eveningStatus");
    }
    statusElement.innerHTML = statusText;
    statusElement.className = statusClass;
    message.innerHTML = "Collection status updated successfully!";
    message.className = "success";
}
```

OUTPUT:




Schedule Waste Pickup
Home
Schedule Pickup
Complaint
Pickup Request
Resident Name
Enter your name
Address
Enter your address
Waste Category
Select Category
Pickup Date
mm / dd / yyyy
Schedule Pickup
© 2026 Waste Collection Management System


Report Uncollected Waste
Home
Schedule Pickup
Complaint
Complaint Form
Resident Name
Enter your name
Address
Enter waste collection location
Complaint Details
Describe the problem
Waste Category
Select Category
Submit Complaint